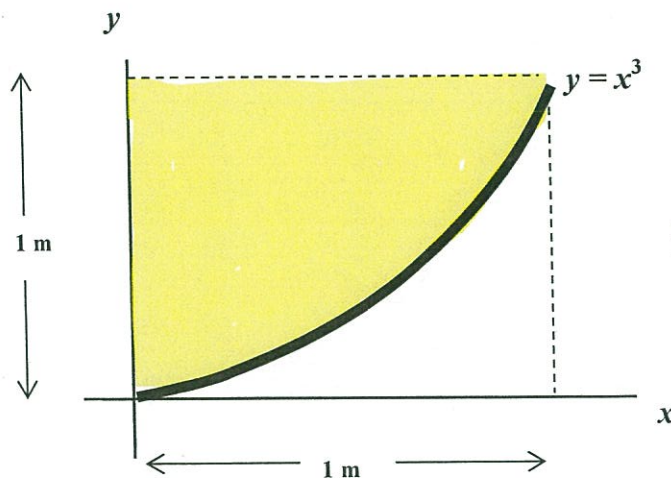


WORKED SOLUTIONS

ENST2.6: CENTROIDS

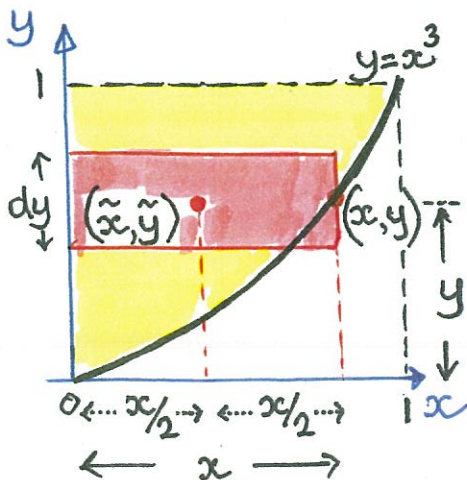
Question 1 Calculate the centroid $(\bar{x}, \bar{y})$ of the shaded area. (Hibbeler, R.C, 2010. *Statics*, Pearson)



Centroid of an area

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA} \quad \bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA}$$

Worked Solution 1



A horizontal strip of area $dA = x dy$ is chosen as the calculations are easier.
[A vertical strip would be $dA = \left(\frac{1-y}{2} + y\right) dx$]

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA} = \frac{\int_0^1 \left(\frac{x}{2}\right) (x dy)}{\int_0^1 (x dy)}$$

where $\tilde{x} = x/2$

Since $y = x^3$, $x = y^{1/3}$. Substituting into above for x :

$$\bar{x} = \frac{\int_0^1 \left(\frac{y^{1/3}}{2}\right) y^{1/3} dy}{\int_0^1 y^{1/3} dy} = \frac{\frac{1}{2} \int_0^1 y^{2/3} dy}{\int_0^1 y^{1/3} dy} = \frac{\frac{3}{10} [y^{5/3}]_0^1}{\frac{3}{4} [y^{4/3}]_0^1} = \frac{3/10}{3/4} = 0.4$$

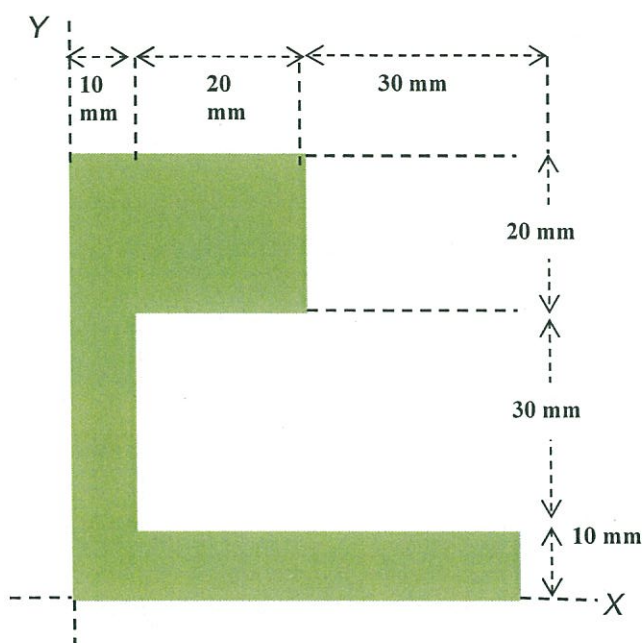
$$\bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA} = \frac{\int_0^1 y (x dy)}{\int_0^1 (x dy)} \quad \text{where } \tilde{y} = y$$

$$\bar{y} = \frac{\int_0^1 y \cdot y^{1/3} dy}{\int_0^1 y^{1/3} dy} = \frac{\int_0^1 y^{4/3} dy}{\int_0^1 y^{1/3} dy} = \frac{\frac{3}{7} [y^{7/3}]_0^1}{\frac{3}{4} [y^{4/3}]_0^1} = \frac{3/7}{3/4} = 0.57$$

The coordinates of the centroid are (0.4, 0.57)

Question 2 Locate the centroid of the composite area shown below with respect to the X- and Y-axes.

(Ivanoff, V. 2010. *Engineering Mechanics*, McGraw Hill)



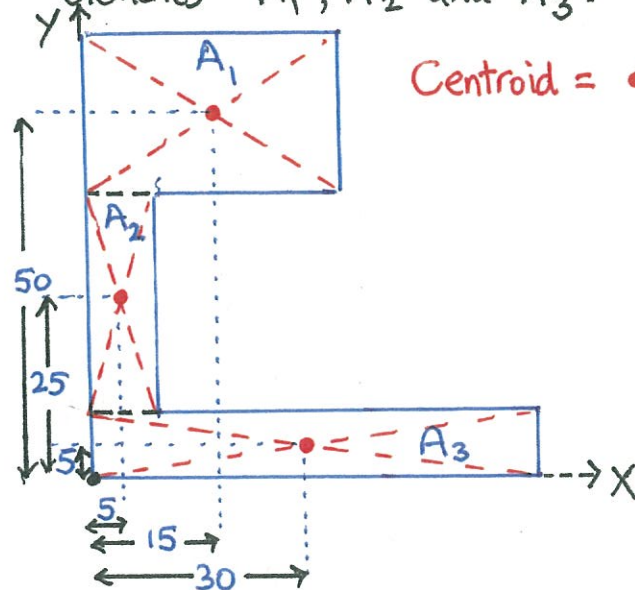
Centroid of a composite area

$$\bar{x} = \frac{\sum (Ax)}{\sum (A)} \quad \bar{y} = \frac{\sum (Ay)}{\sum (A)}$$

where x = horizontal distance to centroids A_1, A_2, A_3 from Y-axis
 y = vertical distance to centroids A_1, A_2, A_3 from X-axis

Worked Solution 2

Divide area into 3 rectangular elements A_1, A_2 and A_3 .



Centroid = •

Element	Area A	Distance		Area Moment	
		x	y	Ax	Ay
1	600	15	50	9 000	30 000
2	300	5	25	1 500	7 500
3	600	30	5	18 000	3 000
$\Sigma =$	1 500	-	-	28 500	40 500

$$\bar{x} = \frac{\sum (Ax)}{\sum (A)} = \frac{28500}{1500} = 19 \text{ mm}$$

$$\bar{y} = \frac{\sum (Ay)}{\sum (A)} = \frac{40500}{1500} = 27 \text{ mm}$$

The coordinates of the centroid are (19mm, 27mm)